CV

Core Competences

Scientific leadership · Project management · Molecular modelling & simulation · Structure prediction · Cheminformatics · Machine learning for chemistry · High-performance computing (HPC) · Student supervision & mentoring · Interdisciplinary collaboration · Scientific communication

Professional and Academic Career

Since 10/2025 — Marie Skłodowska-Curie postdoctoral researcher *Technical University of Munich, Germany* Main project: Machine learning–based compound identification with mass spectrometry

- Developed machine learning models for mass spectrometry signal prediction and dataset similarity analysis.
- Develop protocols for simulating mass spectrometry data for atmospheric compounds.
- Coordinated interdisciplinary projects and supervised students.

9/2022 – 9/2025 — Postdoctoral researcher *Aalto University, Finland* Main project: Machine learning–based compound identification with mass spectrometry

- Developed machine learning models for mass spectrometry signal prediction and dataset similarity analysis.
- Designed molecular descriptors enabling interpretable machine learning models.
- Benchmarked models and descriptors for reaction rate prediction.
- Coordinated interdisciplinary projects and supervised students.

9/2017 – 5/2022 — Early-stage researcher (PhD) *Chalmers University of Technology, Sweden* Main project: Kinetic modeling and molecular structure prediction in polymerization reactions

- Applied steered molecular dynamics, density functional theory, umbrella sampling, and metadynamics for reaction pathway exploration and free-energy profiling.
- Predicted crystal structures of molecular co-crystals and identified plausible reaction products from kinetics/thermodynamics.
- Coordinated multi-site collaborations on crystal structure prediction and lipid conformer analysis; advised students.

9/2024 – 9/2025 — Visiting postdoctoral researcher *University of Gothenburg, Sweden* Simulated mass spectrometry signals using machine learning models, molecular dynamics, reaction exploration and quantum chemistry.

Academic Studies

Period	Degree
9/2017 – 5/2022	**PhD in Chemistry** (Theoretical chemistry), Chalmers University of Technology, Sweden. Award date 02/06/2022. Thesis: *Nitriles in Prebiotic Chemistry and Astrobiology.* Supervisor: Prof. Martin Rahm.
8/2012 – 9/2017	**MEng in Chemical engineering with engineering physics**, Chalmers University of Technology, Sweden. Award date 08/11/2017.
8/2015 – 9/2017	**MSc in Engineering physics** (Nanotechnology master program, integrated), Chalmers University of Technology, Sweden. Award date 08/11/2017.
8/2012 – 6/2015	**BSc in Chemical engineering with engineering physics** (integrated), Chalmers University of Technology, Sweden. Award date 12/06/2015.

Software and Modeling Skills

Programming: Python, MATLAB, Bash — Well Experienced

Machine Learning and Cheminformatics: Scikit-learn, TensorFlow, RDKit, OpenBabel, ASE — Experienced

Molecular Dynamics and Simulation: CP2K, GROMACS, PLUMED — Expert; xTB, QCxMS, VMD — Experienced

High-Performance Computing (HPC): Parallel computing, cluster resource management — Experienced

Version Control: Git — Experienced

Peer-Reviewed Scientific Publications

(15 peer-reviewed articles, 6 first author. Total citations: 110, h-index: 5, i10-index: 4)

1. Cappelletti, M., **Sandström, H.**, & Rahm, M. *ACS Central Science*, 12, 111–121 (2026). DOI: 10.1021/acscentsci.5c01497.
2. Lind, L., **Sandström, H.**, & Rinke, P. *The Journal of Chemical Physics*, 164 (2026). DOI: 10.1063/5.0308548.
3. Madan, I., Aliabadi, S. A., Huhtasaari, J., Matic, E., Hogedal, E., Kamińska, K., Nilsson, F., Stark, A., Izquierdo-Ruiz, F., **Sandström, H.**, Rahm, M. *QRB Discovery*, 6, e23 (2025). DOI: 10.1017/qrd.2025.10012. *[Supervised students and co-created workflow for testing stability of polymers.]*
4. Brean, J., Bortolussi, F., Rowell, A., Beddows, D. C. S., Weinhold, K., Mettke, P., Merkel, M., Kumar, A., Barua, S., Iyer, S., Karppinen, A., **Sandström, H.**, Rinke, P., et al. *ACS ES&T Air*, 2, 1704–1713 (2025). DOI: 10.1021/acsestair.5c00119. *[Supervised PhD student F. Bortolussi in developing and evaluating the machine learning model and workflow.]*
5. Izquierdo-Ruiz, F., Cable, M. L., Hodyss, R., Vu, T. H., **Sandström, H.**, Lobato, A., & Rahm, M. *Proc. Natl. Acad. Sci. U.S.A.*, 122, e2507522122 (2025). DOI: 10.1073/pnas.2507522122. *[Developed and tested crystal structure prediction program workflow for molecular cocrystals.]*
6. Valiev, R. R., Nasibullin, R. T., **Sandström, H.**, Rinke, P., Puolamäki, K., & Kurten, T. *Physical Chemistry Chemical Physics*, 27, 14804–14814 (2025). DOI: 10.1039/D5CP01101A. *[Co-advisor for ML workflow; developed MBTR model.]*
7. Bortolussi, F., **Sandström, H.**, Partovi, F., Mikkilä, J., Rinke, P., & Rissanen, M. *Atmospheric Chemistry and Physics*, 25, 685–704 (2025). DOI: 10.5194/acp-25-685-2025. *[Co-designed study, advised, and contributed to programming and model testing.]*
8. Malaska, M. J., **Sandström, H.**, Hofmann, A. E., Hodyss, R., Rensmo, L., van der Meulen, M., Rahm, M., Cable, M. L., & Lunine, J. I. *Astrobiology*, 25, 367–389 (2025). DOI: 10.1089/ast.2024.0125. *[Performed geometry optimizations, conformer search and student supervision.]*
9. **Sandström, H.**, & Rinke, P. *Geoscientific Model Development*, 18, 2701–2724 (2025). DOI: 10.5194/gmd-18-2701-2025.
10. **Sandström, H.**, Rissanen, M., Rousu, J., & Rinke, P. *Advanced Science*, 11, 2306235 (2024). DOI: 10.1002/advs.202306235.
11. **Sandström, H.**, Izquierdo-Ruiz, F., Cappelletti, M., Dogan, R., Sharma, S., Bailey, C., & Rahm, M. *ACS Earth and Space Chemistry*, 8, 1272–1280 (2024). DOI: 10.1021/acsearthspacechem.4c00088.
12. **Sandström, H.**, & Rahm, M. *The Journal of Physical Chemistry A*, 127, 4503–4510 (2023). DOI: 10.1021/acs.jpca.3c01504.
13. **Sandström, H.**, & Rahm, M. *ACS Earth and Space Chemistry*, 5, 2152–2159 (2021). DOI: 10.1021/acsearthspacechem.1c00195.
14. **Sandström, H.**, & Rahm, M. *Science Advances*, 6, eaax0272 (2020). DOI: 10.1126/sciadv.aax0272.
15. Lindblom, A., Sriram, K. K., Müller, V., Öz, R., **Sandström, H.**, Åhrén, C., Westerlund, F., & Karami, N. *Diagnostic Microbiology and Infectious Disease*, 93, 380–385 (2019). DOI: 10.1016/j.diagmicrobio.2018.10.014. *[Performed fluorescence microscopy assays where I stained, trapped, and photographed plasmids in nanochannels.]*

Teaching and Supervision

Lectures and Exercises

Year	Subject	Degree	Type	Hours/week
2018–2020	Quantum engineering	1st year MSc Nanotechnology	Computer labs	2
2018–2021	Physical chemistry	2nd year BSc Biotechnology	Tutorials and experimental labs	12
2018–2021	Theoretical chemistry	3rd year BSc Chemical engineering	Computer labs	4
2017–2018	Chemistry and biochemistry	1st year BSc Chemical engineering	Experimental labs	8
2014	Calculus	1st year BSc Chemical engineering	Exercise	1

Pedagogical Training

2019 — Teaching, learning and evaluation at Chalmers University of Technology (3 ECTS)

Supervision of Students

Period	Role
SINCE 2024	Supervisor of MSc student at Aalto University
SINCE 2024	Advisor of PhD student at Aalto University
SINCE 2022	Co-supervisor of PhD student at University of Helsinki
11/2024 – 5/2024	Supervisor of BSc student at Aalto University
5/2021 – 9/2021	Co-supervisor of 2 visiting and 3 BSc students at Chalmers University of Technology
1/2021 – 6/2021	Co-supervisor of 6 BSc students at Chalmers University of Technology
6/2020 – 8/2020	Supervisor of 2 BSc students at Chalmers University of Technology
1/2020 – 6/2020	Supervisor of 6 BSc students at Chalmers University of Technology
4/2019 – 7/2019	Supervisor of visiting BSc students at Chalmers University of Technology
4/2018 – 6/2018	Supervisor of one BSc student at Chalmers University of Technology

Funding and Resource Acquisitions

Year	Award
2025	202k EUR Marie Skłodowska-Curie postdoctoral fellowship
2024–2025	LUMI extreme scale access resource allocation
2018–2021	Selected travel grants: Nils Philblad Foundation (2021), Karl and Annie Leon's Foundation (2018–2019)

Conference Contributions

Year	Contribution
2026	**Invited talk** — Network on Mathematical Data Science for Materials Science, Workshop on the Interface of Mathematics and Machine Learning for Applications in Materials Science, University of Glasgow, UK. *Data-driven compound identification with atmospheric mass spectrometry*
2026	**Contributed talk** — Condensed Matter Concepts and Simulation Conference (CCSC 2026), Munich, Germany. *Towards atmospheric compound identification using simulated electron ionization mass spectra*
2025	**Keynote** — Atmospheric day, Sweden. *CLOUDMAP – Advanced identification of atmospheric compounds*
2025	**Invited talk** — Nordic Workshop on AI for Climate Change, Sweden. *Machine learning for atmospheric mass spectrometry*
2024	**Invited talk** — FysKemDagarna (Physics and Chemistry Days), Sweden. *AI in Chemistry: Solving experimental challenges with artificial intelligence*
2023	International Aerosol Modeling Algorithms Conference, USA. *Characterizing Atmospheric Molecules for Machine Learning*
2023	European Aerosol Conference, Spain. *Characterizing Atmospheric Molecules for Machine Learning*
2023	Physics Days, Finland. *Characterizing atmospheric molecules for machine learning*
2022	AbSciCon, USA. *Untangling hydrogen cyanide polymerization using quantum chemistry*

Academic Services

Year	Service
2026	Thesis reviewer and opponent — Universitat Autònoma de Barcelona (UAB), Barcelona, Spain
2025	Reviewer for *ACS Earth Space Chem*, *ACS Omega* and *Atmospheric Chemistry and Physics*
2025	Organizing committee, Nordic Workshop on AI for Climate Change, Sweden
2025	Core member, organizer and Finland representative, Climate AI Nordics Network
2024	Panelist on AI in chemistry, physics, and education, FysKemDagarna
2023	Organizer of workshop hands-on session, Shaking Up Tech. Workshop for underrepresented groups in STEM, Aalto University, Finland
2023	Session chair and organizer, ESTML, Levi, Finland
2022	Session chair, AbSciCon, USA

Languages

Swedish (Excellent) · English (Excellent) · Italian (Intermediate) · French (Basic)